

# E-COMMERCE SALES & CUSTOMER ANALYTICS

## Power BI Data Analyst Project

---

### 1. Project Overview

This project analyzes e-commerce transaction data to understand **sales performance, customer behavior, product performance, payment preferences and regional trends**. The objective was to transform raw transaction data into an **interactive Power BI dashboard** that helps business stakeholders monitor performance and identify areas for improvement.

#### **Dataset**

- 500 Orders
  - 220 Unique Customers
  - 10 Data Fields
  - Transaction period: January–December 2025
- Tools Used:** Excel | Power Query | Power BI | DAX

### 2. Business Problem

The business had transactional data but needed a clear view of its performance.

The analysis was designed to answer:

- How are sales and orders performing over time?
- Which product categories generate the most sales?
- Which payment methods do customers prefer?
- Which states generate the highest sales?
- How many customers are new vs returning?
- Which products are the top revenue contributors?
- Which categories have better customer ratings?

### 3. Data Analyst Approach

I followed an end-to-end data analysis process:

#### **a. Data Understanding**

Reviewed the available transaction fields including order, customer, product, payment, sales, rating and date information.

#### **b. Data Cleaning & Preparation**

- Checked data types
- Validated transaction records
- Reviewed missing/inconsistent values
- Standardized date and numerical fields
- Prepared the dataset for analysis

**c. Data Analysis**

Analyzed the data across:

- Time
- Product category
- Product
- Payment method
- State
- Customer type
- Customer ratings

**d. Power BI & DAX**

Created calculated measures to generate business KPIs and enable dynamic analysis.

**e. Dashboard Development**

Built an interactive dashboard with KPIs, charts, customer analysis and filters.

**f. Insight Generation**

Converted the analysis into business insights and recommendations.

**4. Key KPIs**

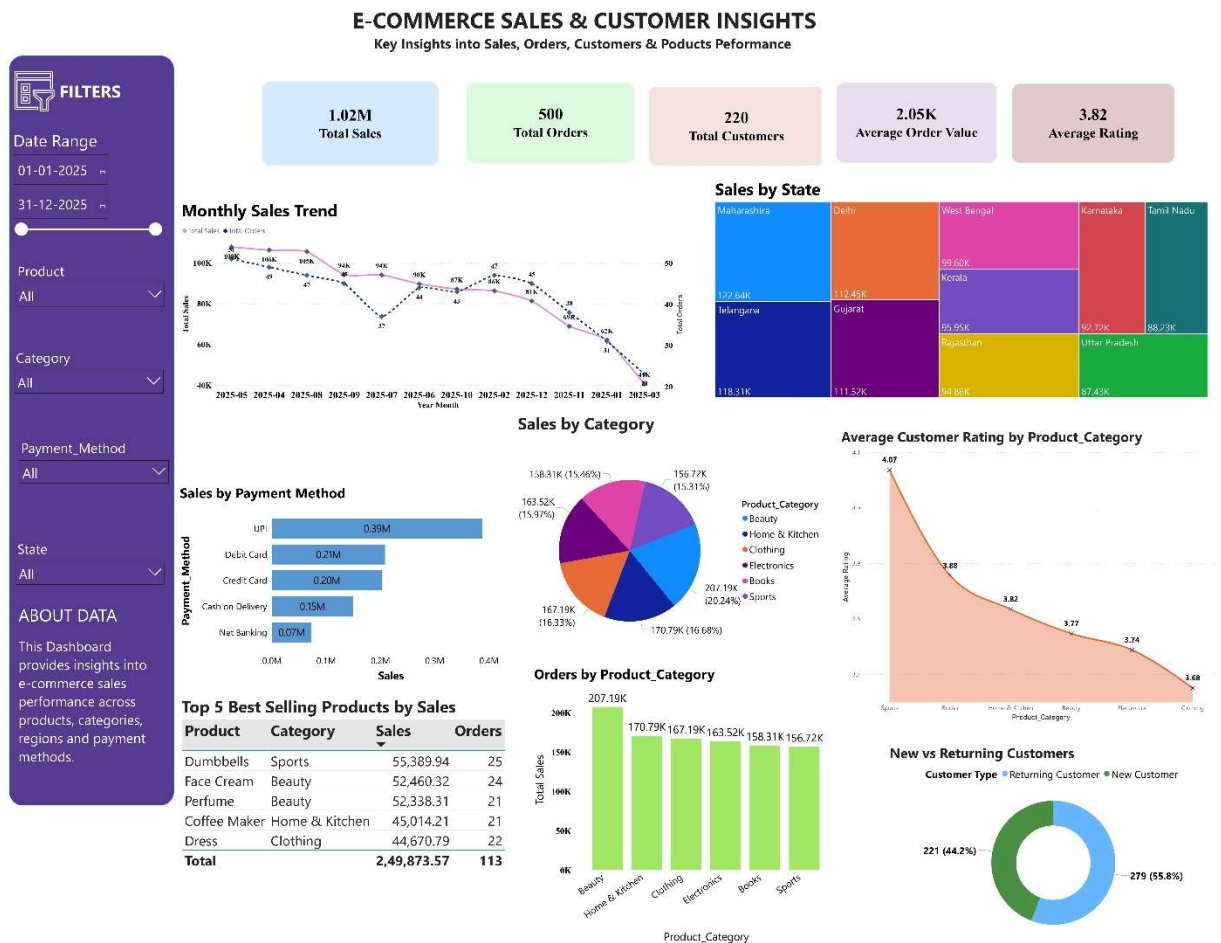
| KPI                     | Result |
|-------------------------|--------|
| Total Orders            | 500    |
| Total Sales             | ₹1.02M |
| Total Customers         | 220    |
| Average Order Value     | ₹2.05K |
| Average Customer Rating | 3.82   |

These KPIs provide a quick overview of the overall e-commerce business performance.

**5. Power BI Dashboard**

**Dashboard Includes**

- Monthly Sales & Order Trend
- Sales by Product Category
- Sales by Payment Method
- Sales by State
- Average Customer Rating by Category
- New vs Returning Customer Analysis
- Top 5 Best-Selling Products
- Interactive filters for Product, Category, Payment Method, State and Date Range



The dashboard was designed to allow business users to interactively explore sales and customer performance instead of relying only on static reports.

## 6. Key Business Insights

- i. **Beauty was the highest-performing category**  
Beauty generated approximately ₹207.19K, contributing around 20.2% of total sales.
- ii. **UPI was the leading payment method**  
UPI generated approximately ₹388K in sales, making it the highest-contributing payment method.
- iii. **Maharashtra had the highest sales**  
Maharashtra generated approximately ₹122.64K, followed by Telangana and Delhi.
- iv. **Sports had the highest customer rating**  
Sports recorded the highest average rating at approximately 4.07.
- v. **Books had the lowest average rating**  
Books recorded an average rating of approximately 3.68, indicating an area that could be investigated further.

vi. **Top products contributed significantly to sales**

The top five products generated approximately **₹249.87K** in sales.

The leading products included:

- Dumbbells
- Face Cream
- Perfume
- Coffee Maker
- Dress

vii. **Sales varied across months**

The monthly trend showed fluctuations in sales and order volumes throughout 2025, allowing the business to identify stronger and weaker sales periods.

## 7. **Business Recommendations**

Based on the analysis, the following actions can be considered:

**Focus on high-performing categories**

Increase promotions and cross-selling opportunities for strong categories such as Beauty.

**Strengthen digital payment experience**

Since UPI generated the highest sales, the business can continue optimizing the UPI checkout experience and digital-payment offers.

**Target high-performing regions**

Regional marketing campaigns can be prioritized in markets such as Maharashtra, Telangana and Delhi.

**Investigate lower-rated categories**

Customer feedback for lower-rated categories such as Books can be analyzed further to identify potential product or customer-experience issues.

**Promote top-selling products**

High-performing products can be prioritized for promotions, recommendations and inventory planning.

**Improve customer retention**

Loyalty offers and personalized recommendations can be used to encourage repeat purchases.

## 8. **Key DAX Measures**

A few important DAX measures used in the dashboard were:

**Total Sales**

Total Sales = SUM(Ecommerce[Order\_Value])

**Total Customers**

Total Customers = DISTINCTCOUNT(Ecommerce[Customer\_ID])

### **Average Order Value**

Average Order Value = DIVIDE([Total Sales], [Total Orders])

### **Average Rating**

Average Rating = AVERAGE(Ecommerce[Customer\_Rating])

These measures were used to create the key business KPIs and support the dashboard analysis.

## **9. Project Outcome**

The project transformed raw e-commerce transaction data into an **interactive business intelligence dashboard**.

It demonstrates practical skills in:

**Data Cleaning → Data Preparation → Data Analysis → DAX → Power BI → Data Visualization → Business Insights**

The dashboard enables stakeholders to quickly understand sales performance, customer behaviour, product performance and regional trends, supporting more data-driven decision-making.